

principles into practice

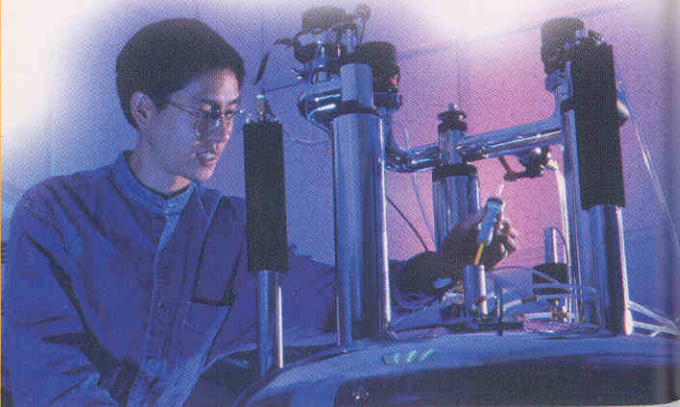


Modern electronic computers already depend on the quantum properties of electrons and atoms for the way their microchips work. But teams around the world are already working on the possibility of a true quantum computer, which would be as much a development from today's best computers as today's computers are a development from the abacus. The best way to picture how such a device would work is in

terms of the idea of a superposition of states. Physicists still argue about whether an entity such as a cat can be in a superposition of states, but there is no doubt that a single electron can exist in this simultaneous duality. For example, an electron in an atom might be able to exist in either of two states—the ground state of lowest energy, or an excited state that it can jump to if it has the energy. These states could correspond to the numbers 0 and 1 of the standard binary code of computers. Such a single

in the lead
The computer and the abacus are worlds apart in terms of their power, but the quantum computer will be an even greater step forward.

the near future
A vial containing quantum computer molecules is loaded into a nuclear magnetic resonance apparatus, where a program of radio-frequency pulses will direct the molecules to carry out a calculation.



quantum unit of information is sometimes called a quantum dot. If light of exactly the right wavelength is shone on the atom for precisely the correct length of time, it is possible to create a situation where there is a 50/50 chance of the electron being in either of these states. In quantum physics, this is the same as saying that the wave function of the electron is a 50/50 mixture of both states—a superposition of states.



current speed limits

The fact that quantum dots can exist in a superposition of states opens up the possibility of making powerful computers by using a small number of physical components. What matters in determining the power of a computer is the number of on/off switches that make up its memory and determines how long the strings of 0s and 1s can be in the binary code of its programs, its calculations, and its answers. Even with microchip technology, there are limitations on the power of computers because electrons have to travel from one switch to another inside the memory. This takes time, which limits the speed of the computer. The flowing electric currents also generate heat, which limits how large the memory can be in physical terms without melting. Computer scientists are trying to get around these problems by using devices such as laser beams to carry the information, but even this kind of technology is limited compared to the power of quantum computing.



ever smaller
The chip being carried by this ant is more powerful than a 1960s room-sized computer. Quantum systems, however, will allow much greater miniaturization when one electron will carry out the work of millions.

too hot to handle
Miniaturization brings its own problems. With components packed closely together, one of the limits on modern PCs is how hot they get.